PM Case Challenge-Ricardo Magdaleno Morales

5.1

Riskiest assumption: I'm assuming that the affected reason is shared on all merchants somehow, this is quite risky as it changes all the strategy to attack the problem because of the impact.

Experiment: Compare the estimated GMV opportunity across merchant GMV tiers and test whether the top 20% of merchants account for a disproportionate share of the opportunity.

Evidence: If I found out that the hypothesis I assumed has shown evidence that is wrong or it's not quite precise then I would change my mind to rethink the discovery, for example if in the experiment it results positive the 20% of merchants as a disproportionate share, i would reject the hypothesis and prioritize a targeted solution.

5.2

Drop off reasons (only 128,914 widget shown):

1. Widget not interaction: 65,675 (50.9%)
2. Payment not selected: 24,067 (18.7%)
3. Other reasons (KYC, underwriting and confirmation): 9,913 (7.7%)

29,259 completed checkout (22.7%)

funnel_step_rea	*drop_off_reason*	COUNTA de ses
completed		28637
	confirmation	192
	declined	216
	kyc	214
Total completed		29259
confirmation	confirmation	2815
Total confirmation		2815
kyc	kyc	5189
Total kyc		5189
underwriting	declined	1909
Total underwriting		1909
widget_intera	payment_metho	24067
Total widget_interacted		24067
widget_show	widget_no_intera	65675
Total widget_shown		65675
Suma total		**128914**

Data issues:

- The widget funnel step is reached but the widget is not always showing and much less is interacted with, so how do we know if it's really reached or if there is a bug? Is this funnel step really needed? for this exercise i excluded this flag from funnel calculations.

- 27 merchants have sandbox_passed after production_credentials so this can have an inconsistency on the data and the expected flow.
- There are merchants that have went_live on the integration but are flagged as churned, this can be normal but we need to implement the churn date flag to review if we need to disconnect or to review if there are checkout sessions after this date that affect the GMV.
- Platform names are not homologated, this can cause a data miss interpretation because you will analyze 12 platforms instead of 4.
- There are 622 sessions that have a drop-off reason but they are marked as completed.

CUT: WooCommerce has a conversion issue on mobile, it has an interaction of 27.1% and a completion rate of 12.2% vs desktop that it has 54% of interactions and completion of 26.5%.

COUNTA de session_id		*funnel_step_reached*						
platform	*device*	completed	confirmation	kyc	underwriting	widget_interacte	widget_shown	Suma total
Custom API	desktop	26.17%	2.63%	2.92%	2.07%	20.11%	46.10%	100.00%
	mobile	22.37%	2.36%	5.17%	1.60%	19.08%	49.43%	100.00%
Total Custom API		23.54%	2.44%	4.47%	1.75%	19.40%	48.40%	100.00%
Shopify	desktop	26.35%	2.71%	2.67%	1.71%	20.48%	46.08%	100.00%
	mobile	22.92%	2.07%	4.95%	1.46%	19.54%	49.06%	100.00%
Total Shopify		23.99%	2.27%	4.24%	1.53%	19.83%	48.13%	100.00%
VTEX	desktop	26.21%	2.69%	2.86%	1.68%	19.94%	46.63%	100.00%
	mobile	23.30%	2.15%	4.96%	1.24%	19.55%	48.80%	100.00%
Total VTEX		24.20%	2.32%	4.31%	1.37%	19.67%	48.12%	100.00%
WooCommer	desktop	26.47%	2.41%	2.63%	1.59%	20.90%	45.99%	100.00%
	mobile	12.21%	0.98%	2.45%	0.89%	10.52%	72.95%	100.00%
Total WooCommerce		16.59%	1.42%	2.51%	1.11%	13.72%	64.66%	100.00%
Suma total		**22.70%**	**2.18%**	**4.03%**	**1.48%**	**18.67%**	**50.94%**	**100.00%**

5.3

My system ranks the merchants with the following variables:

Business Impact: we take in count the tier GMV, GMV relative loss and GMV total relative to calculate it.

Severity: We take in count the average stage where the users drop off, non-completion rate, widget not shown and integration errors.

Confidence: We take in count the volume of sessions and merchant status.

Scalability: We take in count how many merchants have almost a similar issue based on platform, integration, device and drop-off reason.

Decision Rules:

- High GMV´+ High Confidence + Low Scalability → prioritize it if opportunity is materially concentrated
- Low GMV + High Severity + High Scalability → Prioritize it if the same fix applies to a meaningful merchant cohort.
- High GMV + Low Confidence → Do not automatically prioritize it, validate first.

5.4

https://github.com/ricmagmor-hue/Aplazo-GMV

5.5

First I would listen about the other initiative that is competing with mine to understand the business impact and implications because I don't wanna be stubborn with my project if there is another one with higher priority justified.

If i consider my project has more impact i would discuss it with engineering team with arguments such as:

“My ranking takes in consideration GMV impact, repeatability, severity, confidence, scalability and a learning loop so there are many variables that are involved so the decisions are a combination of them to expect the biggest impact to the company or merchant that enhances through time with more data. It's not only a project of my own that wants to be pushed.

This solution also reduces the time needed from KAMs to analyze and decide what to do, also this tool will be in a continuous enhancement with their inputs based on the experience so it will be evolving to be a better tool over time; it’s not only a dashboard for status”

If the solution takes 3 months to be developed then I would like to understand why it takes this time before deciding.
If there is a specific component of the solution maybe we can phase it to start giving business some value ASAP and completing it over time with the rest of the phases.

If even the MVP or phase 1 requires 3 months of development I would re-evaluate it against the potential outcome is big enough vs opportunity cost of the other initiatives in the backlog. If during development we found out that the diagnostic was not clear enough or any of the variables had an opportunity, then I would change my mind to stop it as the evidence is clear enough for it and give the backlog space for another initiative instead of wasting time on something that has been proved wrong.

5.6

- High GMV value but low scalability, this can sound awesome when you mention the $$ but if it’s only adaptable to some merchants it will be risky.
- Custom merchants, this also can sound attractive but as the name says it they have custom plugins so it may only apply to that specific merchant unless the GMV opportunity is large enough to justify its lack of scalability.
- Merchants that have high ranking but have been decreasing their sessions lately, this is because maybe the merchants are near to a churn so first we should align with them in the near future before any development starts.

5.7

Engineering: I would like to understand the reasons you found so we can discuss if you are right, my tool has taken multiple variables that points that this initiative should take the first priority but if i missed something that you noticed i would like to know it so i can adjust it and validate the new priority that is needed.
KAMs: We can prioritize the merchant that you want, at the end the decision is yours, but i want to emphasize that the tool has multiple variables behind that point that the merchant we selected is the most optimal to achieve a better business impact.

Leadership: Is there a reason that I don't acknowledge that supports this ask; is the merchant selected going to churn or disconnect? The tool has the capacity to change the prioritization balance between GMV and scalability so we can move this and select a different strategy.

I'm flexible on the solution, scope and sequencing but, I would not be flexible on the change of any of this without evidence that justifies it.

5.8

ROADMAP

Next 30 days:

Feed the tool to demonstrate that it is well configured and ready to give realistic recommendations, for this we will need to:

- Homologate the data to reduce risks in the analysis.
- Complete the data with more variables such as churn date or disconnection date to increase the confidence of the tool.
- Define cohort benchmarks.
- Release the first ranking so we can align with KAMs if this makes sense or if we need to adjust any variable.
- Run the Experiment mentioned to validate or discard hypothesis.
- Compare recommendations vs KAM judgement.

Next Q':

The tool becomes a trusted input into merchant prioritization, to achieve it:

- Release the initiative prioritized.
- Automate the data feed with the DATA team.
- Let the KAMs input their comments, suggestions and impact of actions so the tool can contrast the real impact vs the estimate it did and adjust for the next initiatives proposed.

Next 6 months:

The tool has demonstrated that it takes the best decisions and is a complement for KAMs instead of a competition:

- Automate the KAMs inputs and impact measures.
- Adjust/increase the ranking variables because the market moves constantly and the tool needs to adapt.
- Implement the integration to the tool so we can detect if there is any issue on the failures/errors of each integration instead of waiting for a ticket from the merchant.

ADOPTION

We need to have clear 2 things, the change management with the tool developed and second the KAM involvement since the requirement because they have the basis of the As Is and what are they missing to enhance their decision making.

It's super important how we manage the change, we need to manage the tool as a complement for KAMs and not a replacement for them.

So, if they are involved in raising the requirement then they will understand that this is a tool to reduce some workload from their end.

To gain trust from the users then we will need to first explain how does the tool works and then compare the decisions that the KAMs may do vs the tool's so they can contrast and give feedback, also if we are showing or giving more information for a decision making then we should also make emphasis on this so they can acknowledge this exists and is adding value to their process.
To measure the usability i would use:

- Decisions taken with tool information (KAM input).
- Business impact done after a decision taken from tool recommendation.
- Override rate + quality, if the KAM decides to go with his own decision and it's better we should measure this as this will help the system to learn for a future.
- I would not use log on KAM login or clicks as they can only login and navigate through it but without taking decisions with the tool.

OBSERVABILITY
Validate the dashboard and movements it does whenever the data has inputs, so we can identify if it's well calibrated.
Contrast information from the past:

1. Input data from the past
2. Request prioritizations for the tool and contrast it to the one that the KAM did in that time before the tool existed and compare results with:
    a. Potential GMV captured vs real GMV obtained with KAM decision.
    b. Incremental GMV, how much GMV of difference is between the potential decision of the tool and the one the KAM decided.
    c. Conversion incremental vs real conversion incremental.
    d. Gaps that KAM didn't identify for any kind of reason.

If this meets all the conditions then we can be sure that the tool is working properly.

FAILURE MODES

- High GMV merchants prioritization, if even when the tool we have bias for the biggest merchants we have a problem; we need to adjust the scalability.
- Low confidence prioritization, if the tools rank higher merchants that have contradictory data we have an issue; we will need to adjust the confidence variable.
- When we introduce small samples of data they rank top, this will be a failure because it shows that the model overreacts; install a rule that for a merchant to enter the ranking you will need a minimum of sessions.
- Lifecycle fail, if any churned merchant appears on any date after churned its a failure reason; flag the merchants churned date and alert if any data is corrupted.

In summary the goal is not to build a smarter dashboard. Is to implement a decision supported system that gets better overtime with more data and with KAM (users) feedback. It will only be successful if the data after the evaluation period is concrete on the results, that it's demonstrated that it generates better outcomes and business impacts than human judgement.